

How Let's Talk Finance 2 is Organized

The 20 topics in **Let's Talk Finance 2** are organized around four levels at which money operates — from the household, through the company, to markets and, finally, to money's role in society.

Part 1: Personal Finance and Household Money

How individuals and households experience inflation, debt, saving, housing and financial fraud.

Part 2: Companies, Work and Money

How companies raise money, how work and pay are changing, and how banks can fail.

Part 3: Markets, Risk and the Global Economy

What moves markets, commodities and currencies, and how developing economies manage risk.

Part 4: Money, Society and the Future

How governments tax and borrow, how money supports culture and philanthropy, and where money itself is heading.

Contents

Part 1 Personal Finance and Household Money	1
1.1. The Cost of Living and Inflation	2
1.2. Debt: Credit Cards, Mortgages and Loans	4
1.3. Saving and Investing for the Long Term	6
1.4. Housing Markets and Affordability	8
1.5. Scams, Fraud and Financial Self-Defence	10
Part 2 Companies, Work and Money	13
2.1. How Companies Raise Money	14
2.2. Startups, Venture Capital and Failure	16
2.3. The Gig Economy and Income Security	18
2.4. Executive Pay and Inequality Inside Firms	20
2.5. Banks: What They Do and How They Fail	22
Part 3 Markets, Risk and the Global Economy	26
3.1. What Moves Stock Markets	27
3.2. Bubbles, Crashes and Manias	29
3.3. Commodities: Oil, Metals and Food	31
3.4. Currencies and Exchange Rates	33
3.5. Emerging Markets and Development Finance	35
Part 4 Money, Society and the Future	38
4.1. Tax: How Governments Raise Money	39
4.2. Government Debt and Deficits	41
4.3. The Business of Sport, Art and Culture	44
4.4. Philanthropy, Foundations and Impact	47
4.5. The Future of Money	49

Part 1

Personal Finance and Household Money

Personal Finance and Household Money starts where finance is most immediately felt: in a household's own budget. Inflation and the cost of living set the backdrop for everything else in this part, since a sustained rise in prices quietly erodes savings and wages alike unless a central bank's target keeps it in check, as Japan's decades of near-zero inflation and Argentina's 211% crisis show from opposite extremes. Debt follows naturally, from the everyday compounding on an unpaid credit-card balance to the newer, less regulated pull of buy-now-pay-later, while saving and investing for the long term asks the flip side of the same question: how compounding, given enough time, works in a saver's favour instead.

Housing markets and affordability bring these threads together in the single largest purchase most households ever make, where interest rates, supply constraints and local policy all interact to decide whether a place to live is treated as shelter or as an investment. The part closes with scams, fraud and financial self-defence, a reminder that every one of these systems, credit, savings, housing, depends on trust, and that trust is exactly what organised fraud is designed to exploit. Across all five topics, the same lesson recurs: household financial wellbeing depends on forces, like inflation and interest rates, that no individual controls, and skills, like recognising a scam, that every individual can still learn.

1.1. The Cost of Living and Inflation

1.1.1. Goal

Explain what inflation is, why it rose around the world after 2021, and why central banks aim for a small positive rate rather than zero.

1.1.2. Reading

Inflation is the rate at which prices rise over time. If a basket of goods costs 3% more than it did a year ago, inflation is 3%. A low, steady rate is normal. The problem comes when the rate jumps, because wages and savings usually do not keep up.

After 2021, prices rose sharply almost everywhere. In the United States, consumer prices in June 2022 were 9.1% higher than a year earlier, the largest 12-month increase in about 40 years^[1]. Petrol was up 59.9% over the year and food was up 10.4%^[1]. In the euro area, inflation reached 10.6% a few months later^[6]. The main causes were global: supply chains were still recovering from the pandemic, and energy and food prices rose sharply after Russia's invasion of Ukraine^[7]. In response, central banks in the United States, the euro area, the United Kingdom and many other countries raised interest rates quickly through 2022 and 2023^[8].

The size of the shock still varied a lot by country. Japan had lived with near-zero inflation, and sometimes falling prices, for about 25 years. There, inflation rising above 2% was itself news. In 2024 the Bank of Japan raised its main interest rate above zero for the first time since 2007^[2]. Argentina was at the other extreme, with annual inflation of 211% in 2023 after years of economic crisis and a large fall in the value of its currency^[3]. Most countries were somewhere in between.

Local factors can add to the global picture. In Japan, a weaker yen made imported food and fuel more expensive. A poor rice harvest in extreme heat then

pushed the price of a 5-kilogram bag of rice to about JPY 4,260 in May 2025, roughly double a year earlier^[4]. A sudden jump in one staple food hits some households hard even when the overall inflation rate looks moderate.

Why aim for 2% rather than 0%? The European Central Bank, like the Bank of England and the US Federal Reserve, targets 2% over the medium term^[5]. A small positive rate gives a safety margin against deflation, which is falling prices. Deflation is dangerous because people delay spending when they expect things to get cheaper, and there is a limit to how far interest rates can be cut to fix it. Japan's long struggle to escape deflation is often given as the clearest example of that risk. When inflation instead runs well above target, a central bank usually raises interest rates. That makes mortgages, business loans and government debt more expensive, and slows the economy — the basic cost of bringing prices back under control.

1.1.3. Vocabulary Focus

inflation, deflation, central bank, inflation target, interest rate, purchasing power

1.1.4. Reading Questions

1. In your own words, what is inflation, and what is deflation?
2. What were the main global causes of the price rises after 2021?
3. How did the experience of inflation differ between Japan and Argentina?
4. How did local factors add to inflation in Japan?
5. According to the reading, why do central banks aim for 2% rather than 0%?

1.1.5. Discussion Questions

1. Which is worse for ordinary people: prices rising quickly, or prices slowly falling? Explain, using an example.
2. Inflation affects savers, borrowers, workers and pensioners differently. Rank these four groups from most harmed to least harmed by a sudden rise in inflation, and explain your order.
3. Should a central bank raise interest rates quickly to stop inflation, even if it causes job losses, or move slowly? Give your view.
4. A government wants to help with the cost of living by sending money to households. How might this help, and how might it make inflation worse?
5. Advise someone on how to protect the value of their savings during a period of high inflation. What are their options, and what is the drawback of each?

1.1.6. Source Notes

1. U.S. Bureau of Labor Statistics: Consumer prices up 9.1 percent over the year ended June 2022, largest increase in 40 years <https://www.bls.gov/opub/ted/2022/consumer-prices-up-9-1-percent-over-the-year-ended-june-2022-largest-increase-in-40-years.htm>
2. CNBC: Bank of Japan ends the world's only negative rates regime in a historic move, abandons yield curve control, March 19, 2024 <https://www.cnbc.com/2024/03/19/bank-of-japan-boj-march-2024-policy-decision-mpm-meeting.html>
3. VOA Learning English: Argentina's Inflation Rises 211.4 Percent, January 2024 <https://learningenglish.voanews.com/a/argentina-s-inflation-rises-211-4-percent/7437750.html>
4. Nippon.com: Soaring Rice Prices in Japan Led to Massive Spike in Private Imports in 2025 <https://www.nippon.com/en/japan-data/h02811/>
5. European Central Bank: Two per cent inflation target <https://www.ecb.europa.eu/mopo/strategy/pricestab/html/index.en.html>
6. Eurostat: Annual inflation up to 10.6% in the euro area, 17 November 2022 <https://ec.europa.eu/eurostat/web/products-euro-indicators/-/2-17112022-ap>
7. Federal Reserve Bank of San Francisco: Global Supply Chain Pressures and U.S. Inflation, June 2023 <https://www.frbsf.org/research-and-insights/publications/economic-letter/2023/06/global-supply-chain-pressures-and-u-s-inflation/>
8. Federal Reserve Bank of Richmond: A Rate Cycle Unlike Any Other, 2023 https://www.richmondfed.org/publications/research/economic_brief/2023/eb_23-26

1.2. Debt: Credit Cards, Mortgages and Loans

1.2.1. Goal

Explain how consumer credit works, why the interest rate varies so much between loan types, and what over-indebtedness looks like.

1.2.2. Reading

In the United States in 2026, the average credit card charged about 21% a year in interest. The average new home loan charged about 6.7%^[1][2]. Both are loans. The difference in price comes mainly from security. A mortgage is backed by the house, which the lender can take and sell if payments stop, so the lender's risk is lower. A credit card is backed by nothing, so the rate is far higher. A car loan, backed by the car, sits in between.

Interest is the price of borrowing, charged as a percentage of the amount still owed. On a credit card, any unpaid interest is added to the balance and then charged interest itself the next month. This is compounding. It is why a balance that is never fully cleared can grow quickly even when no new spending is added to it.

Borrowing has been rising. US households owed US\$18.8 trillion in total in mid-2026, including US\$1.26 trillion on credit cards^[3]. Some countries carry much heavier household debt than others relative to the size of their economy. In South Korea it passed 2,000 trillion won in 2026, and the country now has one of the highest household-debt burdens in the world^[4]. Much of that is home loans. Whether a debt level is dangerous depends on incomes, on interest rates, and on how much of the debt is secured against property that could be sold.

A newer form of borrowing is “buy now, pay later”, which splits a purchase into a few interest-free instalments at the checkout. It grew fast with little regulation. In the year to May 2024, about one in

five UK adults had used it, and the UK market had grown from under GBP 0.1 billion in 2017 to more than GBP 13 billion^[5]. Regulators then moved in. From July 2026 these firms became regulated in the UK and must check that a customer can afford the repayments before offering the plan^[5].

Japan has usually carried lighter household debt than the US or UK. Most Japanese cardholders also pay their balance in full the next month rather than borrowing on it. The defining reform came in 2006, after years of problems with people holding loans from many lenders at once. Before then, consumer-finance firms could charge between 20% and 29% a year, exploiting a gap between two laws^[6]. The reform closed the gap: the maximum interest rate on unsecured consumer loans was cut to 20%, and total borrowing from non-bank lenders was capped at one-third of a person's annual income^[6]. The number of people servicing loans from several lenders fell sharply in the years that followed.

1.2.3. Vocabulary Focus

consumer credit, compounding, secured loan, unsecured loan, over-indebtedness, buy-now-pay-later

1.2.4. Reading Questions

1. Why does a mortgage usually cost much less in interest than a credit card?
2. In your own words, how does compounding make a credit-card balance grow?
3. What does the reading say about how household debt levels compare between countries?
4. How is “buy now, pay later” different from a credit card, and how did regulators respond to it?
5. What did Japan’s 2006 reform change about consumer lending?

1.2.5. Discussion Questions

1. People sometimes describe a mortgage or a student loan as “good debt” and credit-card debt as “bad debt”. Is that a useful distinction? Explain.
2. “Buy now, pay later” is interest-free if you pay on time. Does that make it safe? Give your reasons.
3. Japan capped consumer loan interest at 20% and limited total borrowing to one-third of income. What are the benefits and the drawbacks of rules like these?
4. Rank these four by how much financial trouble they signal: a large mortgage, a car loan, a credit-card balance carried every month, several “buy now, pay later” plans at once. Explain your order.
5. Advise a young adult who has just been offered their first credit card. What would you tell them to do and not do?

1.2.6. Source Notes

1. Federal Reserve (US): Consumer Credit — G.19, August 2026
<https://www.federalreserve.gov/releases/g19/current/g19.pdf>
2. Freddie Mac (US): Primary Mortgage Market Survey <https://www.freddiemac.com/pmms>
3. Federal Reserve Bank of New York: Quarterly Report on Household Debt and Credit, Q2 2026 (data PDF)
https://www.newyorkfed.org/medialibrary/interactives/householdcredit/data/pdf/HHDC_2026Q2
4. Seoul Economic Daily: Korea’s household debt tops 2,000 trillion won for first time (Bank of Korea Q2 2026 data), 19 August 2026 <https://en.sedaily.com/finance/2026/08/19/koreas-household-debt-tops-2000-trillion-won-for-first-time>
5. Financial Conduct Authority (UK): New protections confirmed for Buy Now Pay Later borrowers, 11 February 2026 <https://www.fca.org.uk/news/press-releases/new-protections-confirmed-buy-now-pay-later-borrowers>
6. Federal Reserve Bank of San Francisco: Japan’s New Consumer Finance Law (Asia Focus), February 2007
<https://www.frbsf.org/wp-content/uploads/AsiaFocus-Feb07.pdf>

1.3. Saving and Investing for the Long Term

1.3.1. Goal

Explain the difference between saving and investing, why time and compounding matter, and why low-cost index funds are hard to beat.

1.3.2. Reading

People often use “saving” and “investing” to mean the same thing, but they are different. Saving is keeping money safe and available, usually in cash or a bank deposit, for short-term needs. Investing is putting money into assets such as company shares. Their value can fall as well as rise, but over long periods it is expected to grow more than cash would.

Over the very long run, shares have paid more. One widely used study of 35 markets found that global shares returned an average of about 5% a year after inflation from 1900 to 2024. Government bonds returned under 2%^[1]. That gap compounds, because returns earn further returns. Over several decades, a higher yearly rate produces a far larger final sum. The catch is that the average hides large swings. Any single 20-year period can be much better or much worse than 5%. The same study found the global real return over 2000 to 2024 was only about 3.5% a year. A long-run average is not a promise about the years ahead.

This is why the time horizon matters. Money needed next year should be saved, not invested, because a fall in value cannot be waited out in time. Money not needed for decades can be invested, because the ups and downs tend to even out over a long period. Diversification helps too: spreading money across many companies means no single failure is severe.

For most people, a low-cost “index fund” is the practical choice. An index fund simply tracks a whole market instead of trying to pick winners. Data from S&P Dow Jones Indices shows that over

the 15 years to mid-2025, about 88% of professionally managed US funds investing in large companies did worse than the market index they were measured against^[2]. The fees those funds charge are a large part of the reason. A fund that beats the market in one period often falls behind in the next.

Governments encourage long-term investing through tax-advantaged accounts. Money held inside them grows tax-free or with tax delayed: the ISA in the United Kingdom, the 401(k) and IRA in the United States, compulsory superannuation in Australia. Japan reformed its version, the NISA, in January 2024. It sharply raised the tax-free limits and made the tax break permanent. Take-up was fast, and NISA accounts reached about 27 million by mid-2025^[3]. The push matters because Japanese households have long held about half their financial assets in cash and deposits, far more than households in the United States or the euro area^[4]. That money earned very little as inflation returned after decades of near-zero price rises.

1.3.3. Vocabulary Focus

investing, diversification, index fund, actively managed fund, time horizon, tax-advantaged account

1.3.4. Reading Questions

1. What is the difference between saving and investing, as the reading describes it?
2. What did the long-run study find about the returns on shares compared with government bonds?
3. Why does the reading say money needed next year should be saved rather than invested?
4. What did the S&P Dow Jones data show about professionally managed funds?
5. Why does the reading say the NISA reform “matters” for Japanese households?

1.3.5. Discussion Questions

1. Shares have paid more than cash over the long run, but with large swings along the way. Would you accept those swings for a higher expected return? Explain.
2. About 88% of professional US fund managers did worse than the index. Why might professionals, on average, fail to beat a simple market fund?
3. Compare keeping savings in a bank deposit with investing them in an index fund, for someone aged 30 and for someone aged 65.
4. Governments use tax breaks to encourage long-term investing. Is that a good use of public money? Give your view.
5. Advise someone in their twenties who has just started saving each month. How should they split the money between cash and investments, and why?

1.3.6. Source Notes

1. UBS / Cambridge Judge Business School: Global Investment Returns Yearbook 2025 (Dimson, Marsh, Staunton) <https://www.jbs.cam.ac.uk/2025/report-stocks-have-far-outperformed-over-the-past-125-years/>
2. S&P Dow Jones Indices: SPIVA U.S. Scorecard, Mid-Year 2025 <https://d1e00ek4ebabms.cloudfront.net/production/uploaded-files/spiva-us-mid-year-2025-bc7a7f61-4b27-48b0-b20a-856cc87521d0.pdf>
3. Financial Services Agency (Japan): NISA usage survey (accounts as of end-June 2025) <https://www.fsa.go.jp/policy/nisa/20250924.html>
4. Bank of Japan: Flow of Funds — Overview of Japan, the United States, and the Euro Area, 29 August 2025 <https://www.boj.or.jp/en/statistics/sj/sjhiq.pdf>

1.4. Housing Markets and Affordability

1.4.1. Goal

Explain what drives house prices, why housing became less affordable across much of the rich world, and why homes lose value in some places.

1.4.2. Reading

Does a house always gain value? In much of the world the answer has been yes. But it is not true everywhere, and not in every decade.

House prices are set by demand meeting supply. Demand rises with population, with incomes, and with cheap credit. When interest rates are low, buyers can borrow more, so they bid prices up. Supply is slow to respond, because land is limited and new building is controlled by planning rules. When demand grows faster than supply, prices climb faster than incomes, and housing becomes less affordable.

That is what happened across much of the rich world after the 1990s. In New Zealand, house prices after inflation more than doubled between 2000 and 2007^[1]. Then interest rates rose sharply in 2022, and several of these markets fell. New Zealand prices dropped about 14% from their 2021 peak, and the largest cities fell by around 20%^[1]. Even after those falls, real house prices across advanced economies were still about 20% above their level at the end of the 2008–09 financial crisis^[2].

Governments have tried different levers. Some cities have relaxed their planning rules to allow more building. After Auckland loosened its zoning in 2016, research found the number of new homes approved each year roughly doubled, and rents there rose more slowly than in the rest of New Zealand^[3]. Other governments have tried to cap rents directly. Berlin brought in a rent cap in 2020, but Germany's Federal Constitutional Court

struck it down the next year, ruling that the city had no power to set it^[4]. A few, such as Singapore, have gone further and built most of the housing themselves: about 80% of resident households there live in flats built by a government agency^[5].

Japan is the clearest exception to “houses always go up”. There, the building and the land are valued separately, and the structure is treated as a wasting asset. The tax system assigns a wooden house a life of about 22 years. Older houses are often priced as if the building is worth almost nothing, with nearly all the value in the land. Two things reinforce this. One is heavy new construction: around 800,000 homes were started in the 2023 financial year, despite a shrinking population^[6]. The other is light planning rules that make it easy to knock a house down and rebuild. For a buyer, this means a home is closer to a car than to a store of wealth: it wears out, and most of the lasting value is in the land beneath it. The result is a rising number of empty homes. Japan's 2023 survey counted about 9 million, a record 13.8% of the total housing stock^[7].

1.4.3. Vocabulary Focus

affordability, housing supply, zoning, upzoning, rent control, depreciation

1.4.4. Reading Questions

1. Why do low interest rates tend to push house prices up?
2. Why is the supply of housing “slow to respond” to rising demand?
3. What happened to house prices in New Zealand after interest rates rose in 2022?
4. What was the effect of Auckland loosening its zoning rules in 2016?
5. Why does the reading say Japan is “the clearest exception to ‘houses always go up’”?

1.4.5. Discussion Questions

1. Is a home better understood as an investment or as shelter? Does the answer change depending on the country?
2. Auckland allowed more building; Berlin tried to cap rents. Which approach is more likely to make housing affordable, and why?
3. In Japan, houses lose value while the land holds it. What are the advantages and disadvantages of that pattern for ordinary buyers?
4. Rank these by how much each would improve housing affordability in a high-price city: lower interest rates, looser zoning, a rent cap, a tax on empty homes. Explain your order.
5. Advise a young couple deciding whether to buy a home now or keep renting and invest their savings instead. What should they weigh?

1.4.6. Source Notes

1. Reserve Bank of New Zealand: Update on the housing market (Financial Stability Report special topic), November 2024 <https://www.rbnz.govt.nz/hub/publications/financial-stability-report/2024/november-special-topic-1/update-on-the-housing-market>
2. Bank for International Settlements: Residential property prices statistics (RPP) <https://data.bis.org/topics/RPP>
3. Greenaway-McGrevy, R. & Phillips, P. C. B.: The Impact of Upzoning on Housing Construction in Auckland (Cowles Foundation Discussion Paper 1863) <https://cowles.yale.edu/sites/default/files/2024-02/p1863.pdf>
4. Bundesverfassungsgericht (German Federal Constitutional Court): Act Governing the Rent Cap for Residential Premises in Berlin is void (press release, decision of 25 March 2021) <https://www.bundesverfassungsgericht.de/SharedDocs/Pressemitteilungen/EN/2021/bvg21-028.html>
5. Government of Singapore: Evolution of public housing in Singapore, 30 December 2024 <https://www.gov.sg/explainers/evolution-of-public-housing-in-singapore>
6. Ministry of Land, Infrastructure, Transport and Tourism (Japan): Housing Starts Statistics, FY2023 (via e-Stat) <https://www.e-stat.go.jp/en/stat-search/files?page=1&layout=dataset&toukei=00600120>
7. Nippon.com (citing Statistics Bureau, Ministry of Internal Affairs and Communications): Number of Vacant Homes in Japan Reaches Record 9 Million, 20 May 2024 <https://www.nippon.com/en/japan-data/h01987/>

1.5. Scams, Fraud and Financial Self-Defence

1.5.1. Goal

Explain how modern financial fraud works, why reported losses understate the problem, and how countries differ in protecting victims.

1.5.2. Reading

People in the United States reported losing US\$12.5 billion to fraud in 2024, about a quarter more than the year before. The largest single category was fake investments^[1]. That figure counts only what was reported. Much fraud is never reported, out of embarrassment or because people do not know where to complain, so the real total is higher.

Fraud today is organised and industrial. Criminals use fake investment platforms, phishing messages that copy a real bank, impersonation calls, and “romance” approaches that build a relationship before asking for money. A growing type combines the two: a fake romance leads into a fake investment, sometimes in cryptocurrency, until the victim’s savings are gone. A common method throughout is the authorised push payment scam. The victim is tricked into sending a bank transfer themselves, which makes it harder to reverse, because the payment was technically approved by the account holder. The United Nations has reported that much of this activity is run from large, guarded compounds in South-East Asia, often staffed by trafficked workers. It estimated that losses to victims across East and South-East Asia ran into tens of billions of US dollars in 2023 alone^[2].

Countries protect victims very differently. From October 2024, the United Kingdom has treated this as a consumer protection matter for banks to fix, not just the victim’s problem. It now requires them to reimburse most victims of authorised push payment fraud, up to GBP 85,000 per claim, with the cost shared be-

tween the sending bank and the receiving bank^[3]. In most other countries, including the United States and Japan, whether a victim gets any money back is far less certain. Some countermeasures do seem to help. In Australia, reported scam losses fell by about 26% in 2024 after coordinated action by government, banks and telecoms firms^[4].

Japan has fought “special fraud” for two decades. These scams often begin with a phone call from someone posing as a relative in trouble, or as a police officer. Losses hit a record JPY 72 billion in 2024, up nearly 60% in a year, and about 45% of victims were aged 75 or over^[5]. A separate and newer category, investment and romance scams that start on social media, added a further JPY 127 billion^[5]. The response has focused on adding friction. Many banks suspend cash-machine transfers for older customers who have not used the function recently. Some cap daily withdrawals. Convenience-store staff are trained to question elderly customers who try to buy large numbers of prepaid cards.

Self-defence comes down to a few habits. Treat any unexpected contact about money as suspicious. Never move money because someone has pressured you to act quickly. And check any request through a phone number or app you found yourself, not one the other person gave you.

1.5.3. Vocabulary Focus

fraud, scam, phishing, authorised push payment, investment fraud, reimbursement

1.5.4. Reading Questions

1. Why does the reading say the US\$12.5 billion figure understates the real scale of fraud?
2. What is an authorised push payment scam, and why is it hard to reverse?
3. How does the United Kingdom's approach to fraud victims differ from that of most other countries?
4. What happened to reported scam losses in Australia in 2024, and why?
5. What kinds of "friction" have Japanese banks and shops added to slow fraud down?

1.5.5. Discussion Questions

1. The UK now makes banks refund most transfer-fraud victims. What are the benefits of that rule, and what problems could it create?
2. Fraud in several countries targets elderly people in particular. Whose responsibility is it to protect them — the banks, the family, the government, or the individual? Explain.
3. Adding "friction" (delays, limits, warnings) reduces fraud but also annoys honest customers. Where should the balance sit?
4. Compare protecting people by refunding victims afterwards with protecting them by stopping the payment beforehand. Which is more effective?
5. Design a short checklist you would give an older relative to help them recognise and resist a scam call.

1.5.6. Source Notes

1. Federal Trade Commission (US): New FTC Data Show a Big Jump in Reported Losses to Fraud to \$12.5 Billion in 2024, 10 March 2025 <https://www.ftc.gov/news-events/news/press-releases/2025/03/new-ftc-data-show-big-jump-reported-losses-fraud-125-billion-2024>
2. UN Office on Drugs and Crime: Transnational Organized Crime and the Convergence of Cyber-Enabled Fraud in South-East Asia, 7 October 2024 https://www.unodc.org/roseap/uploads/documents/Publications/2024/TOC_Convergence_Report_2024.pdf
3. Payment Systems Regulator (UK): PS24/7 — Faster Payments APP scams reimbursement requirement, October 2024 <https://www.psr.org.uk/publications/policy-statements/ps247-faster-payments-app-scams-reimbursement-requirement-confirming-the-maximum-level-of-reimbursement/>
4. National Anti-Scam Centre / ACCC (Australia): Australians better protected as reported scam losses fell by almost 26 per cent, 11 March 2025 <https://www.nasc.gov.au/news/australians-better-protected-as-reported-scam-losses-fell-by-almost-26-per-cent>
5. Nippon.com (citing the National Police Agency): Financial Losses Soar as Cases of Scams Jump in Japan, 17 June 2025 <https://www.nippon.com/en/japan-data/h02424/>

Part 2

Companies, Work and Money

Companies, Work and Money moves from the household to the organisations that employ people and raise capital. How companies raise money lays the foundation, contrasting the public discipline of an IPO with the private flexibility that has let firms like SpaceX stay unlisted for years while still raising billions, and startups and venture capital extends the same theme into the highest-risk end of that spectrum, where most funded companies still fail and a rare few become unicorns. The gig economy and income security shifts from how companies raise money to how they organise work itself, asking what happens to benefits, pensions and job security when employment becomes platform-based and irregular.

Executive pay and inequality inside firms turns to a different fault line, the growing gap between what a top executive earns and what a median employee does, and how shareholders can vote, even if only symbolically, on whether that gap is justified. The part ends with banks: what they do and how they fail, using cases from Silicon Valley Bank to the broader mechanics of deposit insurance to explain why a bank run can turn a solvent institution into a failed one within hours. These five topics together describe modern work and enterprise as a set of trade-offs between risk and reward, control and flexibility, that plays out differently for founders, workers and executives alike.

2.1. How Companies Raise Money

2.1.1. Goal

Explain the two basic ways a company funds itself, the difference between staying private and going public, and why fewer companies list than before.

2.1.2. Reading

In 2025 two American technology companies took opposite paths. Figma sold shares to the public and listed on the New York Stock Exchange, raising about US\$1.2 billion. SpaceX did the reverse: it stayed private and raised money by selling shares to selected investors at a valuation of around US\$400 billion, larger than most listed companies, without ever offering its stock on an exchange^[1]. Both routes lead to the same place, which is cash to run and grow the business, but they involve very different trade-offs.

A company can raise money in two basic ways. It can sell equity, meaning shares of ownership, which never has to be repaid but gives new owners a claim on future profits and a say in decisions. Or it can take on debt, meaning bank loans or bonds, which leaves ownership unchanged but must be repaid with interest whether business is good or bad. Equity itself comes in two forms. Private equity is raised from founders, venture capital funds and buyout firms, and the company stays closed. Public equity is raised through an initial public offering, or IPO, when shares are first sold to the public and then trade freely on a stock exchange.

For decades, going public was the natural goal for a successful company. That has changed in the United States and parts of Europe. The number of American listed companies peaked at more than 8,000 in 1996 and is now roughly half that^[2]. At the same time, a large pool of private capital, estimated at around US\$22 trillion worldwide by 2024, now lets big companies raise billions and delay listing for years^[1].

Public markets still work: in 2024 there were about 1,145 IPOs worldwide, and India was among the busiest markets, with a record 268 new listings on its main exchange raising some US\$19.5 billion^[3].

In Japan and much of continental Europe and Asia, bank lending has long mattered more than share issues. Japanese firms traditionally relied on a “main bank” that lent to them, held their shares and watched over management. That arrangement is now slowly unwinding, as companies and banks sell off the cross-held shares that once locked ownership in place. The Tokyo Stock Exchange reorganised itself in 2022 into three segments, Prime, Standard and Growth^[4]. In 2023 it began pressing listed companies to set out plans to raise their share price, aiming especially at the many firms trading below the value of their own assets^[2]. The mix of equity and debt, and of private and public money, differs from one country to the next. But every company faces the same first question: whose money to take, and what to give up for it.

2.1.3. Vocabulary Focus

equity, debt, initial public offering (IPO), listing, private equity, venture capital

2.1.4. Reading Questions

1. How did Figma and SpaceX raise money in 2025, and how were the two approaches different?
2. What is the difference between raising equity and taking on debt?
3. What has happened to the number of listed companies in the United States since 1996?
4. What is private capital, and how has it changed the choice about going public?
5. What was the traditional role of a “main bank” for a Japanese company?

2.1.5. Discussion Questions

1. Selling equity means giving up part ownership and some control. When is that a price worth paying, and when is it not?
2. Many large companies now stay private for years. What do ordinary savers lose if the most successful firms never list?
3. Bank lending dominates corporate finance in Japan; share markets dominate in the United States. What are the strengths of each system?
4. The Tokyo Stock Exchange is pressing companies to raise their share prices. Should an exchange do that, or is it none of its business? Explain.
5. You advise the founder of a fast-growing company deciding whether to go public. Set out the two main questions they should answer first.

2.1.6. Source Notes

1. Fortune: Inside the \$22 trillion world of private capital, 26 October 2025 <https://fortune.com/2025/10/26/what-is-private-capital-credit-22-trillion-industry-equity-debt/>
2. Harvard Law School Forum on Corporate Governance: Looking Behind the Declining Number of Public Companies, 18 May 2017 <https://corpgov.law.harvard.edu/2017/05/18/looking-behind-the-declining-number-of-public-companies/>
3. DD News (Government of India): Indian stock market leads global IPO boom in 2024, NSE raises record \$19.5 billion, January 2025 <https://ddnews.gov.in/en/indian-stock-market-leads-global-ipo-boom-in-2024-nse-raises-record-19-5-billion/>
4. Japan Exchange Group: Announcement of the Results of Market Segment Selection by Listed Companies, 11 January 2022 <https://www.jpx.co.jp/english/corporate/news/news-releases/0060/20220111-01.html>

2.2. Startups, Venture Capital and Failure

2.2.1. Goal

Explain how young companies raise money in stages, why most of them fail, and what venture investors expect from the few that succeed.

2.2.2. Reading

Of every ten new businesses in the United States, about eight survive their first year, about five reach year five, and about three reach year ten^[1]. Among companies backed by venture capital, the odds of a big payout are longer still: one widely cited study found that roughly three in four never return their investors' cash^[4]. Failure is not a rare accident in this part of finance. It is the normal outcome, and the whole model is built around that fact.

A startup usually raises money in stages, each tied to reaching a goal. The earliest money, often called seed funding, comes from founders, friends and specialist early investors. It buys the company time to build a product and find its first customers. If that works, a venture capital fund may lead a larger round, labelled Series A, then Series B, and so on. Each round buys a slice of ownership, usually at a higher price than the last, and the money is meant to last until the next milestone. Venture capital is money raised from pension funds, universities and wealthy individuals and placed in young, high-risk companies in exchange for equity. The investors know most of their bets will fail, so they need the rare winner to return many times its cost. The end goal, or exit, is a sale to a larger company or a stock market listing, which is how the fund turns its shares back into cash.

The size of the funding pool rises and falls sharply. Global venture investment reached roughly US\$440 billion in 2025, with nearly half going to artificial intelligence companies, after a steep fall in 2022

and 2023^[2]. When funding tightens, companies that raised money at a high valuation must often accept a lower one in the next round, a “down round” that dilutes founders and early staff. The most striking recent failure was WeWork, an office-space company valued at US\$47 billion in early 2019, which then failed to list, lost billions and filed for bankruptcy in 2023^[2].

Japan's venture scene is smaller but growing. One estimate put Japan at seven “unicorns”, meaning private companies worth over US\$1 billion, in 2023, against more than 600 in the United States^[3]. In 2022 the government adopted a five-year plan to lift the number of startups and unicorns sharply, with public money and tax incentives behind it^[5]. Europe and India have both built up sizeable venture markets of their own over the same period. Whether in Tokyo, Bangalore or California, the arithmetic is the same: investors accept many total losses in exchange for a small chance of backing a company that becomes very large. That is why a single success, such as an early stake in a company that later lists, can carry a whole fund.

2.2.3. Vocabulary Focus

startup, seed funding, funding round, down round, exit, unicorn

2.2.4. Reading Questions

1. Of ten new United States businesses, roughly how many reach year five and year ten?
2. What happens at each stage of startup funding, from seed money onward?
3. Why do venture investors need the rare successful company to return many times its cost?
4. What is a “down round”, and who does it hurt?
5. How does the number of “unicorns” in Japan compare with the United States, and what has the Japanese government done?

2.2.5. Discussion Questions

1. In venture capital, most investments are expected to fail. Is that a sensible way to use pension savings? Give reasons.
2. A “down round” punishes founders and early employees for raising money at a high price earlier. Is that fair? Explain.
3. Governments in Japan and elsewhere use public money to encourage startups. When is that a good use of public funds, and when is it not?
4. Almost half of recent venture money went to artificial intelligence companies. What are the risks of investment being concentrated in one field?
5. You advise someone deciding whether to join a startup or take a secure job at a large firm. Set out the two main trade-offs they should weigh.

2.2.6. Source Notes

1. U.S. Bureau of Labor Statistics: Business Employment Dynamics — Entrepreneurship and the U.S. Economy (survival rates of establishments)
<https://www.bls.gov/bdm/entrepreneurship/entrepreneurship.htm>
2. Crunchbase News: North American Startup Funding Soared 46% In 2025, Driven By AI Boom (with global 2025 totals), January 2026 <https://news.crunchbase.com/venture/north-american-startup-funding-2025-data-ai-us-investment/>
3. International Monetary Fund: Riding Unicorns — Startups and Venture Capital in Japan (Working Paper WP/24/246), December 2024 <https://www.imf.org/en/publications/wp/issues/2024/12/06/riding-unicorns-startups-and-venture-capital-in-japan-559169>
4. Harvard Business School: The Venture Capital Secret: 3 Out of 4 Start-Ups Fail, 2012
<https://www.hbs.edu/news/Pages/item.aspx?num=487>
5. Cabinet Secretariat of Japan: Startup Development Five-year Plan, November 2022
https://www.cas.go.jp/jp/seisaku/atarashii_sihonsyugi/pdf/sdfyplan2022en.pdf

2.3. The Gig Economy and Income Security

2.3.1. Goal

Explain what platform work is, why the “employee or self-employed” question matters so much, and how different countries have answered it.

2.3.2. Reading

A delivery rider logs into an app on Friday evening, accepts trips for four hours, and is paid for each one. There is no set shift, no manager, and no promise of work next week. The rider values the freedom to choose hours, but carries risks an employee does not: earnings that swing from week to week, and usually no paid holiday, no sick pay, no employer pension contribution and no unemployment cover. This is platform work, meaning paid tasks arranged through an app or website, and it has grown fast. The International Labour Organization estimates that the number of digital labour platforms worldwide grew about five-fold in the decade to 2020^[1].

The core dispute is a legal one: are these workers really self-employed, as the platforms say, or are they employees entitled to labour protections? The answer decides who bears the cost of holidays, illness and old age. Courts and parliaments have reached different conclusions. In February 2021 the United Kingdom Supreme Court ruled that Uber drivers are “workers”, a middle category that brings the minimum wage and paid leave. It also held that their working time counts from the moment the app is switched on^[2]. The European Union adopted a directive in 2024 that presumes a platform worker is an employee unless the company can show otherwise, with member states to write it into national law by December 2026^[3]. Spain passed a law in 2021 making delivery couriers employees^[5].

The United States has gone the other way, at least in its largest state. California

passed a law in 2019 that would have made many gig drivers employees. Ride-hailing and delivery companies then funded a ballot measure, Proposition 22, that carved themselves out, spending more than US\$200 million on the campaign. California’s Supreme Court upheld the measure in 2024, so drivers there remain independent contractors with a limited set of benefits and no sick pay or unemployment cover^[6]. Most other United States states never adopted a test like California’s in the first place.

Japan has taken two narrower routes rather than one big reclassification. A Freelance Act, in force from November 2024, does not make freelancers employees. Instead it puts duties on the companies that hire them: state the work and the fee in writing, pay within 60 days, guard against harassment, and give 30 days’ notice before ending a continuing contract^[4]. Separately, in 2022 a Tokyo labour board found that Uber Eats couriers are “workers” under union law and can bargain collectively, a decision the company has challenged^[7]. Across all these systems the same tension runs. The flexibility that makes platform work attractive is bound up with the absence of the protections that make an income secure.

2.3.3. Vocabulary Focus

platform work, the gig economy, self-employed, independent contractor, worker status, collective bargaining

2.3.4. Reading Questions

1. What risks does a platform worker carry that an employee does not?
2. What did the United Kingdom Supreme Court decide about Uber drivers in 2021?
3. What does the European Union's 2024 directive presume, and what must member states do by December 2026?
4. How did California end up keeping gig drivers as independent contractors?
5. What does Japan's Freelance Act require of companies that hire freelancers?

2.3.5. Discussion Questions

1. Platform workers often say they value flexibility above employee benefits. Should the law let them make that trade, or protect them regardless? Explain.
2. If gig drivers become employees, companies may offer less flexible work or hire fewer people. How should lawmakers weigh that risk?
3. The United Kingdom uses a three-way split of employee, "worker" and self-employed. Would a middle category help or confuse things in your country?
4. Japan chose fair-dealing rules for freelancers rather than making them employees. What are the strengths and limits of that approach?
5. You advise a government drafting new rules for platform work. Suggest two protections that should apply no matter how workers are classified.

2.3.6. Source Notes

1. International Labour Organization: World Employment and Social Outlook 2021 — The role of digital labour platforms, 23 February 2021 <https://www.ilo.org/resource/news/rapid-growth-digital-economy-calls-coherent-policy-response-0>
2. The Supreme Court of the United Kingdom: Uber BV and others v Aslam and others [2021] UKSC 5, 19 February 2021 <https://www.supremecourt.uk/cases/uksc-2019-0029>
3. European Parliament: Parliament adopts Platform Work Directive, 24 April 2024 <https://www.europarl.europa.eu/news/en/press-room/20240419IPR20584/parliament-adopts-platform-work-directive>
4. Japan Fair Trade Commission: Act on Ensuring Proper Transactions Involving Specified Entrusted Business Operators (tentative translation) https://www.jftc.go.jp/en/legislation_gls/index.html
5. European Agency for Safety and Health at Work (OSHA-EU): Policy Case Study — Spain: The "Riders' Law," 2022 https://osha.europa.eu/sites/default/files/2022-01/Spain_Riders_Law_new_regulation_digital_platform_work.pdf
6. CalMatters: Prop. 22 gig-work law upheld by California Supreme Court, 25 July 2024 <https://calmatters.org/economy/2024/07/prop-22-california-gig-work-law-upheld/>
7. Japan Today: Tokyo says Uber Eats labor union has collective bargaining rights, November 2022 <https://japantoday.com/category/business/tokyo-says-uber-eats-labor-union-has-collective-bargaining-rights>

2.4. Executive Pay and Inequality Inside Firms

2.4.1. Goal

Explain how top executive pay is built, how it compares with a typical worker's pay in different countries, and what rules and votes now surround it.

2.4.2. Reading

In 2024 the chief executives of the 350 largest United States companies were paid about 281 times as much as a typical worker in their industry, up from 21 times in 1965^[1]. The gap is now the subject of laws, disclosures and shareholder votes in most rich countries, and it looks very different from one country to the next.

Top executive pay usually has three parts. There is a base salary, fixed and relatively modest as a share of the total. There is an annual bonus tied to that year's results. And there is stock-based pay, meaning shares or options that vest over several years and now make up the large majority of a big American chief executive's package. Because so much of the pay is in stock, its final value depends on the share price, which is why supporters call it pay for performance and critics call it a windfall in a rising market.

Two kinds of rule have grown up around the question. The first is disclosure. Since 2018, United States listed companies must publish the ratio of chief executive pay to the pay of their median employee, meaning the employee exactly in the middle of the pay range^[4]. The United Kingdom has required a similar figure since 2020^[5]. The second kind of rule is the shareholder vote, known as "say on pay". In the United Kingdom and much of Europe the vote on pay policy is binding, so a company may not pay outside an approved policy. In the United States the vote is only advisory, meaning the board can proceed even if shareholders vote no. Switzerland went furthest, banning sign-on and severance bonuses for listed-com-

pany executives by referendum in 2013^[6]. Votes against pay do happen: Netflix shareholders rejected the company's executive pay package in an advisory vote in 2023^[2].

Japan sits at the other end of the range. The median ratio of executive to employee pay across listed Japanese companies is roughly 10 times, far below the American figure^[3]. Since 2010, any director paid 100 million yen or more must be named individually in the company's annual report. Fewer than 300 executives crossed that line in 2009; a record 1,345 did so at 606 companies in the year to March 2025, and the highest-paid were mostly foreign executives at Japanese firms^[3]. The rise reflects the spread of stock-based pay rather than a jump in salaries. The underlying argument is the same everywhere: is very high pay the price of attracting and motivating rare talent, or is it simply what executives can command when boards decide their rewards?

2.4.3. Vocabulary Focus

executive compensation, base salary, stock-based pay, pay ratio, say-on-pay vote, disclosure

2.4.4. Reading Questions

1. How did the ratio of top United States chief executive pay to a typical worker's pay change between 1965 and 2024?
2. What are the three parts of top executive pay, and which part is usually the largest for a big American chief executive?
3. What must United States and United Kingdom listed companies now disclose about pay?
4. What is the difference between a binding and an advisory "say on pay" vote?
5. How does the executive-to-employee pay ratio in Japan compare with the United States, and what has happened to the number of Japanese executives paid 100 million yen or more?

2.4.5. Discussion Questions

1. Much top pay is in stock, so its value rises and falls with the share price. Does that tie pay to performance, or mostly to luck? Explain.
2. In the United States the "say on pay" vote is only advisory. Should shareholders be able to block a pay package outright? Give reasons.
3. Japan discloses the pay of any director earning 100 million yen or more, by name. Is naming individuals fair, or an invasion of privacy?
4. Switzerland banned sign-on and severance bonuses for executives. Would such a ban help or backfire? Explain.
5. You advise a company board setting the chief executive's pay. Suggest two principles it should follow to keep the package defensible to shareholders.

2.4.6. Source Notes

1. Economic Policy Institute: CEO Pay, September 2025 <https://www.epi.org/publication/ceo-pay/>
2. U.S. Securities and Exchange Commission: Pay Ratio Disclosure (Item 402(u) of Regulation S-K), final rule 2015 <https://www.sec.gov/newsroom/press-releases/2015-160>
3. Nippon.com (data from Tokyo Shoko Research): Foreign Executives Prominent in Japan's Top Earners List, 25 August 2026 <https://www.nippon.com/en/japan-data/h02867/>
4. U.S. Securities and Exchange Commission: Pay Ratio Disclosure — Final Rule, adopted 2015, effective 2018 <https://www.sec.gov/rules/final/2015/33-9877.pdf>
5. UK Government (legislation.gov.uk): The Companies (Miscellaneous Reporting) Regulations 2018 <https://www.legislation.gov.uk/ukxi/2018/860/contents/made>
6. Library of Congress, Global Legal Monitor: Switzerland: Referendum on Executive Pay, 27 March 2013 <https://www.loc.gov/item/global-legal-monitor/2013-03-27/switzerland-referendum-on-executive-pay>

2.5. Banks: What They Do and How They Fail

2.5.1. Goal

Explain the basic job of a bank, why that job makes every bank fragile, and how deposit insurance and a central bank try to prevent runs.

2.5.2. Reading

On Friday 10 March 2023, Silicon Valley Bank in California was closed by regulators. The day before, its customers had tried to withdraw more than US\$40 billion in a single day, faster than any bank run in modern records^[1]. It was, at the time, the second-largest bank failure in United States history^[5]. What went wrong was not fraud or reckless lending. It was the ordinary business of banking, done at the wrong moment.

A bank takes in deposits, which customers can withdraw at any time, and lends most of that money out for years, as mortgages and business loans. Turning short-term deposits into long-term loans is called maturity transformation, and it is useful: it channels idle savings into productive use. But it leaves every bank unable to repay all its depositors at once. If enough customers ask for their money together, even a sound bank can be forced to sell assets at a loss and collapse. That is a bank run, and the fear of one can cause one.

Governments try to break that logic in two ways. The first is deposit insurance: a government-backed promise that small savers will be repaid up to a limit even if their bank fails. In the United States the limit is US\$250,000 per depositor per bank^[2]. In Japan the Deposit Insurance Corporation covers ordinary deposits up to 10 million yen, plus interest, and covers everyday settlement accounts in full^[3]. The point is to remove the reason for an ordinary saver to join a run. The second tool is the central bank acting as lender of last resort, lending cash to a sound bank

against good collateral so it does not have to sell assets in a panic. When the British lender Northern Rock faced a run in 2007, queues formed outside its branches until the government guaranteed its deposits^[6].

Sometimes a bank is judged too important to close. In March 2023 the Swiss authorities arranged for UBS to take over its troubled rival Credit Suisse over a single weekend. Central bank funding and a government guarantee stood behind the deal. Depositors lost nothing, but shareholders and holders of certain risky bonds lost heavily^[4]. Japan faced a broader failure in 1997 and 1998, when Hokkaido Tokai Bank and the Yamaichi securities house collapsed. The government eventually took two large banks into public ownership and injected public money to steady the rest, a slow and costly clean-up that took years^[7]. The lesson from all these cases is that a bank's stability rests less on its own vault than on the belief that the safety net behind it will hold.

2.5.3. Vocabulary Focus

deposit, maturity transformation, bank run, deposit insurance, lender of last resort, collateral

2.5.4. Reading Questions

1. What did customers of Silicon Valley Bank try to do the day before it was closed?
2. What is maturity transformation, and why does it leave every bank unable to repay all depositors at once?
3. What is deposit insurance, and what are the limits in the United States and Japan?
4. What does it mean for a central bank to act as “lender of last resort”?
5. What happened when the Swiss authorities dealt with Credit Suisse in 2023, and who lost money?

2.5.5. Discussion Questions

1. Deposit insurance protects savers but is paid for by the banking industry and, ultimately, could fall on taxpayers. Is that protection worth the cost? Explain.
2. A bank run can bring down a bank that is basically sound. Does that justify governments guaranteeing deposits, or does it encourage banks to take more risk?
3. Some banks are treated as too important to fail. What problems does that create, and what could reduce them?
4. Japan's 1997–98 crisis involved many banks at once, not one. Why is a system-wide banking crisis harder to handle than a single failure?
5. You advise a country setting its deposit-insurance limit. Suggest two things it should weigh in choosing the figure.

2.5.6. Source Notes

1. Board of Governors of the Federal Reserve System: Review of the Federal Reserve's Supervision and Regulation of Silicon Valley Bank, 28 April 2023 <https://www.federalreserve.gov/publications/files/svb-review-20230428.pdf>
2. Federal Deposit Insurance Corporation: Understanding Deposit Insurance <https://www.fdic.gov/resources/deposit-insurance/understanding-deposit-insurance>
3. Deposit Insurance Corporation of Japan: A Guide to the Deposit Insurance System https://www.dic.go.jp/english/e_yokinsha/page_000017.html
4. Swiss Financial Market Supervisory Authority (FINMA): FINMA approves merger of UBS and Credit Suisse, 19 March 2023 <https://www.finma.ch/en/news/2023/03/20230319-mm-cs-ubs/>
5. CNBC: Here's how the second-biggest bank collapse in U.S. history happened in just 48 hours, 10 March 2023 <https://www.cnbc.com/2023/03/10/silicon-valley-bank-collapse-how-it-happened.html>
6. Bank of England: Ten years on — lessons from Northern Rock (speech), 2017 <https://www.bankofengland.co.uk/-/media/boe/files/speech/2017/ten-years-on-lessons-from-northern-rock>
7. Nippon.com: Lessons of the 1997 Financial Crisis in Japan <https://www.nippon.com/en/currents/d00360/>

Part 3

Markets, Risk and the Global Economy

Markets, Risk and the Global Economy zooms out to the forces that move prices and capital across entire economies. What moves stock markets opens by naming the four real drivers behind any market move, earnings, interest rates, sentiment and index concentration, correcting the common habit of assigning a single tidy cause to what is usually several forces acting together. Bubbles, crashes and manias traces the same pattern across four centuries, from Dutch tulip bulbs to Japan's late-1980s property bubble to a 2022 cryptocurrency collapse, showing that cheap money, herd behaviour and the belief that "this time is different" recur in every era.

Commodities, currencies and emerging markets close the part by looking at risk that crosses borders directly: a bakery in Tokyo paying more for flour because of a war fought thousands of kilometres away, a currency's value shifting because of a decision made by a foreign central bank, and a developing economy's access to capital drying up the moment global interest rates rise. These five topics share a single insight: in a genuinely global financial system, a shock rarely stays confined to the market or country where it started.

3.1. What Moves Stock Markets

3.1.1. Goal

Explain the main forces that move stock market indexes — earnings, interest rates, sentiment, and concentration in a few large companies.

3.1.2. Reading

Why did the market fall today? Financial news answers this every evening, usually naming one cause: an interest-rate decision, a disappointing profit report, a change in mood. All three are real drivers, and a fourth has grown more important — how much of an index depends on just a few very large companies. A stock index is a single number tracking a basket of shares, weighted by market capitalisation, the total value of each company's shares.

The most durable driver is company earnings, the actual profits firms make. A share is ultimately a claim on a company's future profits, so a higher expected profit stream is worth more. In 2025 the S&P 500, the main US index, returned about 18%, and more than three-quarters of that came from higher earnings rather than investors paying more per dollar of profit^[1]. When a large company's quarterly report misses expectations, its shares, and the index, can drop the same day. Earnings and interest rates together are what analysts call fundamentals.

Interest rates are the second driver. Higher rates make borrowing costlier for companies and make safe bonds a more attractive alternative to shares, so rate decisions move markets directly. On 18 December 2025 the European Central Bank held its main rate at 2%, its fourth pause in a row^[2]; Europe's Stoxx 600 rose about 1% that session and ended 2025 up roughly 17%^[3]. A US rate cut reaches further still, often sending money toward emerging markets in search of higher returns as the dollar softens.

The third driver is investor sentiment — collective confidence or fear. It is hard to measure, but the closest gauge is the VIX, an index of how much volatility options traders expect in the US market over the next month^[4]. Often called the market's "fear gauge," it stays low when markets are calm and jumps when investors rush to buy protection during a sell-off. A calm market can turn volatile within hours if confidence breaks, with no change in earnings or rates.

The fourth driver is index concentration. By the end of 2025 the ten largest companies in the S&P 500 made up a record 40.7% of its value, roughly double their share a decade earlier, most of them tied to artificial intelligence^[5]. When an index leans this heavily on a few firms, its daily direction can be set by one company's news. Japan shows that the usual rules can still surprise. On 16 June 2026 the Bank of Japan raised its policy rate to 1%, the highest since 1995^[6]. Higher rates are supposed to weigh on shares, but the Nikkei 225 rose instead, passing 70,000 in intraday trading for the first time, because investors read the move as removing uncertainty rather than adding cost^[7]. On any given day, the same news can push a market either way, depending on what investors already expected.

3.1.3. Vocabulary Focus

stock index, market capitalisation, fundamentals, index concentration, investor sentiment, volatility

3.1.4. Reading Questions

1. What four things does the reading say move stock prices?
2. In 2025, did the S&P 500's gain come mainly from higher earnings or from investors paying more per dollar of profit?
3. Why do higher interest rates tend to push share prices down?
4. What does the VIX measure, and how does it behave in a calm market compared with a sell-off?
5. Why did the Nikkei 225 rise after the Bank of Japan raised interest rates in June 2026, when higher rates usually weigh on shares?

3.1.5. Discussion Questions

1. Financial news usually explains a market move with a single cause. Given the four drivers in the reading, how useful are those daily explanations?
2. When ten companies make up 40% of an index, is that index still a good measure of "the market"? Explain.
3. Sentiment can move prices for days without any change in earnings or rates. Should investors pay attention to sentiment, or ignore it? Give reasons.
4. Compare earnings and interest rates as drivers of stock prices. Which would you watch more closely, and why?
5. A friend in Japan asks whether a Bank of Japan rate rise is good or bad for their share investments. Advise them, using the reading.

3.1.6. Source Notes

1. First Trust: "The S&P 500 Index 2025 Recap," 8 January 2026 (2025 total return ~17.9%; over three-quarters from EPS growth, ~2.5 points from P/E expansion)
<https://www.ftportfolios.com/Commentary/EconomicResearch/2026/1/8/the-sp-500-index-2025-recap>
2. European Central Bank: "Monetary policy decisions," 18 December 2025
<https://www.ecb.europa.eu/press/pr/date/2025/html/ecb.mp251218~58b0e415a6.en.html>
3. CNBC: "European markets close: Stoxx 600, FTSE, DAX, CAC," 18 December 2025
<https://www.cnbc.com/2025/12/18/european-markets-on-decpoint18-stoxx-600-ftse-dax-cac.html>
4. S&P Dow Jones Indices: "VIX" (introduction to the Cboe Volatility Index)
<https://www.spglobal.com/spdji/en/vix-intro/>
5. RBC Wealth Management: "The 'Great Narrowing': S&P 500 concentration" (22 January 2026)
<https://www.rbcwealthmanagement.com/en-us/insights/the-great-narrowing-sp-500-concentration>
6. CNBC: "Bank of Japan hikes rates to 1%, highest since 1995," 16 June 2026
<https://www.cnbc.com/2026/06/16/boj-rate-hike-historic-inflation.html>
7. The Japan Times: "Nikkei 225 breaks 70,000 on hopes for peace and after rate increase," 16 June 2026
<https://www.japantimes.co.jp/business/2026/06/16/markets/nikkei-70000-first-time-june/>

3.2. Bubbles, Crashes and Manias

3.2.1. Goal

Show the recurring pattern behind asset bubbles across four centuries, and why the aftermath often matters more than the crash.

3.2.2. Reading

In the first week of February 1637, buyers at a tulip-bulb auction in Haarlem simply stopped bidding. Sellers lowered their prices, then lowered them again, and still no one bought. Within days the market for Dutch tulip bulbs — a single bulb had earlier been traded for a house and farmland — had collapsed^[1]. Tulip mania is the oldest well-known asset bubble, and almost every element of it has re-occurred since.

A bubble forms when the price of an asset rises far above any sensible valuation. Three things carry it up: cheap or borrowed money, called leverage; herd behaviour, people buying because others are buying; and a belief that “this time is different,” so the usual warnings no longer apply. This becomes a speculative mania, buying driven by expected further price rises rather than value. Economists Carmen Reinhart and Kenneth Rogoff traced that “this time is different” belief through financial crises across 66 countries and eight centuries, finding it before nearly every one^[2]. The crash comes when buyers stop believing the price will keep rising.

The 1929 Wall Street crash is the classic modern case. The Dow Jones Industrial Average peaked near 381 in September 1929, then fell for almost three years to a low of 41.22 in July 1932, a peak-to-trough decline of about 89%. It did not regain its 1929 level until November 1954, a quarter of a century later^[3]. The US dot-com crash of 2000 and the 2008 global financial crisis followed the same shape: a boom built on new technology or easy credit, then a long

repair. In each case the warning signs were visible beforehand and widely dismissed.

Japan’s late-1980s bubble is one of the best-documented anywhere. The Nikkei 225 closed at 38,915.87 on 29 December 1989, lifted by cheap credit and soaring land values used as loan collateral^[4], and took 34 years to reclaim that level, in February 2024^[4]. Property fell even harder: commercial land prices in six major cities dropped by roughly 87% from their peak and kept falling for 15 years, not turning up again until 2005^[5]. Banks were left holding loans backed by land worth a fraction of its former value, and the economy grew slowly for two decades.

The pattern reached new assets without changing. The cryptocurrency Luna had traded near US\$120 in early April 2022; over three days in May, after confidence in its linked “stablecoin” broke, it fell from about US\$80 to almost nothing^[6][7]. The details differ each time — bulbs, shares, land, tokens — but the sequence does not: easy money and rising prices draw in more buyers, the belief that the rise cannot stop becomes general, and then it stops. What varies most is the aftermath, and the risk of a slow one grows with the length of the boom and the amount of borrowed money behind it — a few weeks for Luna, more than three decades for the Nikkei.

3.2.3. Vocabulary Focus

asset bubble, speculative mania, herd behaviour, leverage, “this time is different” thinking, peak-to-trough decline

3.2.4. Reading Questions

1. What happened at the Haarlem tulip auction in February 1637?
2. What three things does the reading say carry an asset bubble upward?
3. How long did it take the Dow to regain its 1929 level, and how long did it take the Nikkei 225 to regain its 1989 level?
4. Why were Japanese banks left in a weak position after the late-1980s bubble burst?
5. The reading says “the sequence does not” change. What sequence does it mean?

3.2.5. Discussion Questions

1. The reading says the belief “this time is different” appears before nearly every bubble. Why is that belief so hard to resist during a boom?
2. Bubbles have formed around very different assets: bulbs, shares, land, cryptocurrencies. What makes some assets more prone to bubbles than others?
3. The aftermath of a bubble ranged from weeks (Luna) to decades (the Nikkei). What determines how long the recovery takes?
4. Compare the 1929 crash with the 2022 Luna collapse. What is similar, and what is different?
5. Design three warning signs an investor could watch for to judge whether a rising market has become a bubble.

3.2.6. Source Notes

1. Federal Reserve Bank of New York, Liberty Street Economics: “Crisis Chronicles: Tulip Mania, 1633–37” <https://libertystreeteconomics.newyorkfed.org/2013/09/crisis-chronicles-tulip-mania-1633-37/>
2. Princeton University Press: Carmen M. Reinhart and Kenneth S. Rogoff, “This Time Is Different: Eight Centuries of Financial Folly” (2009) <https://press.princeton.edu/books/paperback/9780691152646/this-time-is-different>
3. Federal Reserve History: “Stock Market Crash of 1929” <https://www.federalreservehistory.org/essays/stock-market-crash-of-1929>
4. Nippon.com: “Nikkei Index Sets First Record High Since 1989” <https://www.nippon.com/en/japan-data/h01927/>
5. Nippon.com: “Heisei Blues: The Post-Bubble Struggles of Japan’s Financial Sector” <https://www.nippon.com/en/in-depth/d00470/heisei-blues-the-post-bubble-struggles-of-japan%E2%80%99s-financial-sector.html>
6. Federal Reserve Bank of Richmond: “Why Stablecoins Fail: An Economist’s Post-Mortem on Terra” (Economic Brief, 2022) https://www.richmondfed.org/publications/research/economic_brief/2022/eb_22-24
7. Harvard Law School Forum on Corporate Governance: “Anatomy of a Run: The Terra Luna Crash” (22 May 2023) <https://corpgov.law.harvard.edu/2023/05/22/anatomy-of-a-run-the-terra-luna-crash/>

3.3. Commodities: Oil, Metals and Food

3.3.1. Goal

Explain why the prices of raw materials swing so sharply, and contrast a sudden supply shock with a slower, structural price story.

3.3.2. Reading

In March 2022, a bakery in Tokyo that had paid a steady price for flour for years received a new quote: about a third higher, with no promise it would hold. Its wheat came from abroad, and two weeks earlier Russia had invaded Ukraine — together the two countries supplied roughly 30% of the world's wheat exports, so the price jumped on every market at once^[1]. The bakery could not negotiate with the war; it could only pay the new world price or use less.

This is the defining feature of commodities — raw materials such as oil, wheat and metals, traded in bulk and largely interchangeable between suppliers. Their prices are set globally, so a shock anywhere reaches buyers everywhere and feeds quickly into inflation. Oil showed it in the same weeks: Brent crude, the main benchmark price for oil, rose above US\$120 a barrel, its highest since 2008^[7]. At the end of March 2022 the US released 180 million barrels from its strategic reserve, the largest drawdown in its history, to ease the shock^[2].

The food side ran in parallel but was fixed differently. The UN Food and Agriculture Organization's Food Price Index hit an all-time high in March 2022, up almost 13% in a month^[1]. Rather than wait for prices to draw out new supply, the UN and Türkiye brokered a shipping corridor. It moved about 33 million tonnes of grain out of Ukraine's Black Sea ports over the following year^[3], and the index fell back: a supply shock can sometimes be reversed by a specific arrangement.

Metals show a different price risk. In March 2022 the London Metal Exchange suspended nickel trading and cancelled about US\$12 billion of trades after the price doubled within hours, driven by supply fears and one producer's large losing bet^[4]. Even a commodity exchange, it turns out, can stop working under stress. For some metals the risk is a government's policy: China mines roughly 60% of the world's rare earths and processes about 90%, with export controls since 2023 on a widening list of critical minerals^[5].

Japan feels all of this more sharply than most, producing almost none of it. It imports nearly all its oil, gas and coal, and its food self-sufficiency rate on a calorie basis was 38% in fiscal 2023, below the government's 45% target^[6]. Too small a share of any market to move the price, Japan is close to a pure price-taker — and in 2022 the weakening yen cut into its purchasing power still further. Its usual defences are the ones every importer uses: long-term contracts, part-ownership of overseas fields, and large stockpiles, including well over 200 days of oil reserves^[8]. Rice is the revealing exception: Japan grows almost all its own, yet a domestic shortage still pushed prices up sharply in 2024^[9]. Producing something yourself lowers exposure to world prices; it does not remove price risk.

3.3.3. Vocabulary Focus

commodity, benchmark price, supply shock, strategic reserve, price-taker, self-sufficiency rate

3.3.4. Reading Questions

1. Why did the Tokyo bakery's flour price jump in March 2022?
2. What does the reading say is the "defining feature" of commodities?
3. How was the 2022 food-price shock partly reversed?
4. Why does Japan feel commodity-price swings more sharply than most countries?
5. Why does the reading call rice "the revealing exception"?

3.3.5. Discussion Questions

1. Governments hold strategic reserves of oil and grain. Should every import-dependent country do this, given the cost of storage? Explain.
2. For some metals, the main price risk is one country's export policy rather than weather or war. What can other countries do about that kind of risk?
3. The reading says producing something yourself lowers exposure to world prices but does not remove price risk. Do you agree? Use the rice example.
4. Compare a sudden oil supply shock with China's slower-moving rare-earth export controls. Which is harder for a government to manage, and why?
5. Advise a food importer on three steps it could take now to reduce the impact of a future commodity-price shock.

3.3.6. Source Notes

1. UN News: "Ukraine war drives international food prices to 'new all-time high'" (FAO Food Price Index, March 2022) <https://news.un.org/en/story/2022/04/1115852>
2. U.S. Department of Energy: "History of SPR Releases" (2022 coordinated release: 180 million barrels, largest in SPR history) <https://www.energy.gov/hgeo/opr/history-spr-releases>
3. Council of the European Union: "Ukrainian grain exports explained" (Black Sea Grain Initiative) <https://www.consilium.europa.eu/en/infographics/ukrainian-grain-exports-explained/>
4. U.S. Office of Financial Research: "Central Clearing and Trade Cancellation: The Case of LME Nickel Contracts on March 8, 2022" <https://www.financialresearch.gov/working-papers/2024/12/10/central-clearing-and-trade-cancellation/>
5. Center for Strategic and International Studies: "The Consequences of China's New Rare Earths Export Restrictions" <https://www.csis.org/analysis/consequences-chinas-new-rare-earths-export-restrictions>
6. The Japan Times: "Japan's food self-sufficiency rate remains at 38%," 10 October 2025 <https://www.japantimes.co.jp/news/2025/10/10/japan/food-self-sufficiency-rate/>
7. U.S. Energy Information Administration: Europe Brent Spot Price FOB <https://www.eia.gov/dnav/pet/hist/rbrtem.htm>
8. U.S. Energy Information Administration: China, the United States, and Japan hold most strategic oil inventories in 2025 <https://www.eia.gov/todayinenergy/detail.php?id=67504>
9. Nippon.com: Rice Shortages in Japan: Ongoing Cuts to Production at Fault <https://www.nippon.com/en/in-depth/d01044/>

3.4. Currencies and Exchange Rates

3.4.1. Goal

Explain what makes a currency rise or fall, and challenge the idea that a weak currency is simply bad news.

3.4.2. Reading

Many people assume a weak currency is always a problem. The story of the Japanese yen between 2022 and 2024 shows why that is too simple. An exchange rate is the price of one currency in another, set like any price by supply and demand: when more people want to hold a currency it strengthens, when they sell it weakens. What changes that demand most is the gap between interest rates in different countries — money moves toward wherever it earns more, so when one central bank raises rates and another does not, the higher-rate country's currency usually gains.

That is exactly what happened to the yen. Through 2022 the US Federal Reserve raised its policy rate sharply to fight inflation, while the Bank of Japan held its rate near zero, and investors moved money out of yen into higher-yielding dollars. The yen fell from about 115 per US dollar at the start of 2022 to around 150 by late 2022, and touched 160 in 2024, its weakest in over three decades^[1]. Japan's Ministry of Finance intervened, selling dollars to buy yen, spending about JPY 9.8 trillion in 2022 to slow the fall^[2] — intervention can smooth a move, but rarely reverses a gap in interest rates.

The effects split the economy in two. A weak yen made Japanese exports cheaper abroad, and large exporters such as car-makers reported record profits, partly because foreign earnings converted into more yen^[3]. But the same weak yen made imported oil, gas and food more expensive, pushing consumer prices up and cutting the real spending power of house-

holds and small firms that buy from abroad. This is the core trade-off of a floating exchange rate — one that markets set freely: it cannot be strong for exporters and strong for importers at the same time.

Not every country lets its currency float. Some run a fixed exchange rate, promising to keep their currency at a set value against another, usually the dollar. Hong Kong has done this since 1983, and several Gulf states peg to the dollar too. A peg brings stability for trade, but it means importing the anchor country's monetary policy: when the Fed raised rates in 2022, Hong Kong's rates had to follow, whether or not its economy needed them^[5]. Other economies manage their rate loosely, intervening only at the edges.

The wider lesson is that an exchange rate is a relative price, not a scoreboard. When the yen was at 160, headlines called it a national weakness, yet Japan's export sector was thriving and tourism hit record numbers because visitors found the country cheap. By mid-2025, as the Bank of Japan raised rates and the Fed began cutting, the gap narrowed and the yen recovered toward 140^[4]. A currency's level is rarely simply good or bad. It shifts value between groups — exporters and importers, producers and consumers, savers and borrowers — and which way you read it depends on which side you are on.

3.4.3. Vocabulary Focus

exchange rate, floating exchange rate, fixed exchange rate, currency intervention, interest rate differential, currency peg

3.4.4. Reading Questions

1. What does the reading say moves currency demand the most?
2. Why did the yen fall so far between 2022 and 2024?
3. What did Japan's Ministry of Finance do in 2022, and what are the limits of that action?
4. How did the weak yen help exporters but hurt households?
5. What does a country give up when it fixes its exchange rate to the dollar?

3.4.5. Discussion Questions

1. The reading argues a weak currency is not simply bad. Do you find that convincing? Give reasons from the text.
2. A fixed exchange rate brings stability but means following another country's interest-rate decisions. When is that trade worth making?
3. Currency intervention "can smooth a move but rarely reverses" an interest-rate gap. If so, why do governments still intervene?
4. Compare the position of a Japanese carmaker and a Japanese food importer during the weak-yen period. Whose situation would you rather manage?
5. You advise a government whose currency is falling fast. Rank these options and justify your order: raise interest rates, intervene in the market, introduce a peg, do nothing.

3.4.6. Source Notes

1. Reuters: "Japanese yen: a timeline of the currency's slide to 34-year lows"
<https://www.reuters.com/markets/currencies/japanese-yen-timeline-currencys-slide-34-year-lows-2024-04-29/>
2. Ministry of Finance, Japan: "Foreign Exchange Intervention Operations"
https://www.mof.go.jp/english/policy/international_policy/reference/feio/index.html
3. The Japan Times: "Weak yen boosts Japanese automakers' profits despite sales challenges"
<https://www.japantimes.co.jp/business/2024/05/09/companies/toyota-honda-earnings-weak-yen/>
4. Reuters: "Yen strengthens as BOJ-Fed policy gap narrows"
<https://www.reuters.com/markets/currencies/yen-strengthens-boj-fed-policy-gap-narrows-2025-07-15/>
5. Hong Kong Monetary Authority: Linked Exchange Rate System <https://www.hkma.gov.hk/eng/key-functions/monetary-stability/linked-exchange-rate-system/>

3.5. Emerging Markets and Development Finance

3.5.1. Goal

Explain what “emerging market” means for an investor and a government, and how developing countries raise money to grow.

3.5.2. Reading

In 2024 two large developing economies borrowed on very different terms. India sold ten-year government bonds at yields close to 7%, with demand strong enough that the country was added to a major global bond index, drawing in billions of new foreign money^[1]. Egypt, the same year, could only secure funding after an expanded IMF programme and a large UAE investment deal, its dollar bonds having traded at distressed levels^[2]. Both are called emerging markets, but the label covers a wide range of risk.

An emerging market is an economy that is growing and opening to foreign investors but is not yet as stable, wealthy or well-regulated as an advanced economy. For investors, these markets offer higher potential returns and faster growth, in exchange for greater currency risk, political risk and the danger of sudden capital outflows — money leaving quickly when sentiment turns. Index providers such as MSCI and FTSE Russell decide which countries qualify, and inclusion matters: when a country joins an index, funds that track it must buy its assets, lowering the government’s borrowing cost.

Developing countries fund growth from three main sources, each with a catch. The first is foreign direct investment — a foreign company building a factory or buying a lasting stake in a local business. It is the most stable form of foreign money because it is hard to pull out quickly, but it flows mainly to a few larger economies. The second is borrowing on international markets in dollars or euros, which is fast to arrange but creates a cur-

rency mismatch: if the local currency falls, the debt becomes far heavier. The third is lending from development finance institutions — bodies such as the World Bank, the regional development banks and national aid agencies — which offer lower rates and longer terms but attach conditions and move slowly.

Japan is one of the largest bilateral providers of this last kind of finance. Through the Japan International Cooperation Agency and its leading role at the Asian Development Bank, it funds transport, energy and water projects across South and Southeast Asia, often as long-term loans in yen at low interest^[3]. Supporters say this builds lasting capacity; critics note it can tie borrowers to Japanese contractors and add to debt loads.

The recurring danger is the sudden stop. Emerging markets depend on foreign capital that can reverse fast when global interest rates rise or risk appetite falls, as it did across 2022 when the US Federal Reserve tightened policy and investors pulled back from riskier assets worldwide^[4]. Countries that had borrowed heavily in dollars, such as Sri Lanka, defaulted; countries with deeper local savings, larger reserves and credible central banks, such as India and Indonesia, absorbed the shock^[5]. The clearest lesson of development finance is that how a country borrows — the currency, the maturity, the lender — often matters more than how much.

3.5.3. Vocabulary Focus

emerging market, capital outflow, foreign direct investment, currency mismatch, development finance institution, sudden stop

3.5.4. Reading Questions

1. How did India's and Egypt's borrowing terms differ in 2024?
2. What does the reading say an emerging market offers investors, and what do they risk in return?
3. Why does joining a global bond index lower a government's borrowing cost?
4. What is the "catch" with borrowing on international markets in dollars or euros?
5. What role does Japan play in development finance, and what do critics say about it?

3.5.5. Discussion Questions

1. The reading says how a country borrows often matters more than how much. Do you agree? Use the examples in the text.
2. Foreign direct investment is the most stable foreign money but flows mainly to larger economies. What could a smaller country do to attract more of it?
3. Development finance institutions offer cheap, long loans but attach conditions and move slowly. Are those trade-offs worth it for a borrowing government?
4. Compare the positions of Sri Lanka and India during the 2022 tightening. What made the difference?
5. You advise the finance ministry of a fast-growing emerging economy. Recommend a borrowing strategy that reduces the risk of a sudden stop, and explain your choices.

3.5.6. Source Notes

1. J.P. Morgan: "India's inclusion in the GBI-EM Global index suite" <https://www.jpmorgan.com/insights/global-research/markets/india-bond-index-inclusion>
2. International Monetary Fund: "IMF Executive Board Completes Review of Egypt's Extended Arrangement" <https://www.imf.org/en/News/Articles/2024/03/29/pr2494-egypt-imf-execboard-completes-first-and-second-reviews-eff-arrangement>
3. Asian Development Bank: "Japan and ADB" (fact sheet) <https://www.adb.org/publications/japan-fund-poverty-reduction-fact-sheet>
4. International Monetary Fund: "Global Financial Stability Report, October 2022" <https://www.imf.org/en/Publications/GFSR/Issues/2022/10/11/global-financial-stability-report-october-2022>
5. Federal Reserve Bank of Dallas: Emerging-market countries insulate themselves from Fed rate hikes, 8 August 2023 <https://www.dallasfed.org/research/economics/2023/0808>

Part 4

Money, Society and the Future

Money, Society and the Future closes the resource by asking what money is for beyond markets and firms. Tax and government debt open the part together, examining how governments raise the money that pays for public services and what happens when spending consistently outruns revenue, a tension that toppled a British prime minister in 2022 and that Japan has so far managed without a crisis for reasons that are still debated. The business of sport, art and culture turns to a less obvious kind of finance, where a single painting can sell for more than an orchestra's annual budget and Saudi Arabia's sovereign wealth fund buys football clubs partly for "soft power" rather than financial return alone.

Philanthropy, foundations and impact examines how large-scale giving is structured, taxed and increasingly judged by measurable outcomes rather than good intentions alone, while the future of money closes the resource by asking where money itself is heading, from Sweden's now-reversing retreat from cash to central bank digital currencies still stuck at the pilot stage almost everywhere. Across all five topics, the resource ends on the same note it began: money is not only a tool for individual households or companies, but a shared system that societies continually choose, and re-choose, how to run.

4.1. Tax: How Governments Raise Money

4.1.1. Goal

Set out the main sources of government revenue, how the mix varies between countries, and who actually bears each tax.

4.1.2. Reading

Where does government money actually come from? Almost every road, school, court and pension is paid for by tax, and the mix of taxes a country chooses decides who carries the cost.

The broadest measure is the tax-to-GDP ratio — total tax revenue as a share of the size of the economy. Across the OECD it averaged about 33.9% in 2023, ranging from about 17.7% in Mexico to about 43.8% in France. In a typical OECD country, social-security contributions and personal income tax each raise roughly a quarter of revenue. Goods-and-services taxes, including VAT, add about a fifth, corporate income tax about a tenth, and property taxes a few percent^[1].

Countries at similar income levels choose very different structures. The United States has no federal VAT and leans heavily on personal income tax and payroll taxes, with sales taxes set by states at relatively low rates^[2]. Most EU countries run high standard VAT rates that raise a large, stable share of revenue, usually the biggest single source after social contributions — neither design is more “normal” than the other.

A country’s tax structure follows its revenue base and shifts when that base looks less secure. Gulf states long funded themselves from oil and charged residents almost nothing; the UAE introduced a 5% VAT in 2018 and a 9% federal corporate tax from June 2023, while still charging no personal income tax^[3].

Tax is also now an area of international coordination. Around 140 countries agreed in 2021 to a 15% minimum effective

tax rate on large multinationals’ profits, so a company shifting profit to a very-low-tax country now has the shortfall “topped up” elsewhere. The rules took effect from 2024 in the EU, UK, Japan and South Korea, though the US moved to withdraw its support in 2025^[4]. Underneath all this sits tax incidence: the person or company that hands money to the government is often not the one who bears the cost. VAT is collected from businesses but paid by final consumers; corporate tax is legally paid by companies but its burden is shared, in disputed proportions, among shareholders, employees and customers.

Japan’s tax story is dominated by one instrument and one problem. The instrument is the consumption tax, introduced at 3% in 1989 and raised in steps to 5% (1997), 8% (2014) and 10% (2019, with food kept at 8%)^[5]. Every increase cost the governing party politically, yet the consumption tax is now Japan’s largest single national tax, because it is the most reliable way to raise revenue from a population with a growing share of retirees. The problem is that an ageing society needs ever more spending on pensions and health care while the number of working-age taxpayers falls, so Japan also leans hard on social-security contributions levied on wages rather than on higher income-tax rates. Even so, tax does not cover what the government spends; the gap is filled by selling bonds, which is the subject of the next topic.

4.1.3. Vocabulary Focus

tax-to-GDP ratio; value-added tax (VAT) / consumption tax; social-security contributions; progressive / regressive tax; tax incidence; global minimum tax ("Pillar Two")

4.1.4. Reading Questions

1. What does the tax-to-GDP ratio measure?
2. How does the United States' tax structure differ from that of most EU countries?
3. Why did the United Arab Emirates introduce a VAT and a corporate tax?
4. What problem is the global minimum tax meant to solve?
5. Why is Japan's consumption tax now its largest single national tax, despite each increase being politically costly?

4.1.5. Discussion Questions

1. VAT takes a larger share of a low earner's income than a high earner's. Does that make it a bad tax, or is it justified by other qualities? Explain.
2. Countries at similar income levels tax very differently. What might explain those choices — history, politics, or something else?
3. The global minimum tax needs many countries to act together, and the US has stepped back. Can it work without full participation?
4. Compare taxing consumption (VAT) and taxing income. Which would you rely on more, and why?
5. You advise a government that must raise revenue equal to 2% of GDP. Recommend which taxes to raise or introduce, and explain who would bear the cost under your plan.

4.1.6. Source Notes

1. OECD: Revenue Statistics 2024 (tax-to-GDP ratios; tax structures)
https://www.oecd.org/en/publications/2024/11/revenue-statistics-2024_6e88b46e.html
2. OECD: Revenue Statistics — United States (country note)
<https://www.oecd.org/content/dam/oecd/en/topics/policy-sub-issues/global-tax-revenues/revenue-statistics-united-states.pdf>
3. UAE Ministry of Finance: "The Ministry of Finance announces the introduction of a Corporate Tax in the UAE" <https://mof.gov.ae/en/news/the-ministry-of-finance-announces-the-introduction-of-a-corporate-tax-in-the-uae/>
4. OECD: "Minimum Tax Implementation Handbook (Pillar Two)"
<https://www.oecd.org/content/dam/oecd/en/topics/policy-sub-issues/global-minimum-tax/minimum-tax-implementation-handbook-pillar-two.pdf>
5. Nippon.com: "The Political History of Japan's Consumption Tax"
<https://www.nippon.com/en/features/h00013/the-political-history-of-japan%E2%80%99s-consumption-tax.html>

4.2. Government Debt and Deficits

4.2.1. Goal

Explain the difference between deficits, debt and debt-to-GDP, and why bond markets, not arithmetic, usually set the limit.

4.2.2. Reading

On 23 September 2022 the UK government announced about GBP 45 billion of tax cuts with no stated plan to pay for them. Government bonds, known as gilts, fell at their fastest rate in decades, the pound hit a record low near US\$1.035, and forced selling by pension funds threatened a wider crash. On 28 September the Bank of England began emergency gilt purchases, authorised up to GBP 65 billion^[1][2]. The finance minister was sacked, the prime minister resigned after 49 days, and almost every measure was reversed^[7].

The episode turns on three linked ideas. A deficit is the gap in one year between what a government spends and what it raises in tax. Debt is the accumulated total of past deficits. The debt-to-GDP ratio compares that stock to a year's national output, because a larger economy can carry more debt. A government can lower the ratio by running surpluses or by growing — or inflating — GDP faster than the debt.

Governments run deficits routinely and can carry high debt for decades, but the bond market sets a shifting, sometimes sudden limit, demanding a higher yield when it doubts a government's plans. That is the market discipline other checks lack. Even the United States has now been downgraded by all three major agencies — S&P in 2011, Fitch in 2023, Moody's in May 2025 — each citing large and growing deficits, with federal debt near US\$40 trillion^[3]. Its bonds remain the world's benchmark safe asset, so it still borrows easily, but the cost of servicing that debt

reached roughly US\$880 billion in fiscal 2024, more than the defence budget^[4].

The euro area tried to prevent crises with rules — a 3%-of-GDP deficit limit and 60%-of-GDP debt limit, set in 1997, suspended during COVID and replaced in 2024 with country-specific spending paths^[5]. Rules, though, meet politics. In 2024 Kenya's government proposed about US\$2.6 billion of new taxes, partly to meet IMF targets, when interest payments alone already consumed about 37% of revenue. Mass protests forced the president to withdraw the bill^[6]. Deficit reduction is a political problem before it is an arithmetic one.

Japan is the puzzle that makes the topic interesting. Its government debt is well over 200% of GDP, comfortably the highest in the rich world, yet its bonds pay very low interest and there has been no run on them^[8]. The debt is held almost entirely by Japanese institutions and households, the Bank of Japan owns around half, and a quarter-century of deflation kept interest rates near zero. So Japan has never faced the market pressure that forces other governments to act — raising the consumption tax is so costly at the ballot box that it has happened only three times in 35 years, even as spending on an ageing population keeps rising^[9]. Japan shows a country can carry extraordinary debt for a long time when the conditions are right — and “the conditions are right” is not a plan.

4.2.3. Vocabulary Focus

budget deficit / surplus; government (sovereign) debt; debt-to-GDP ratio; bond yield; sovereign credit rating; market discipline

4.2.4. Reading Questions

1. What happened to UK government bond prices after the 23 September 2022 announcement, and how did the Bank of England respond?
2. What is the difference between a deficit and debt?
3. What are two ways a government can lower its debt-to-GDP ratio?
4. Why does the United States still borrow easily despite three credit downgrades?
5. Why has Japan been able to carry very high debt without a crisis?

4.2.5. Discussion Questions

1. The reading says “deficit reduction is a political problem before it is an arithmetic one.” Explain what that means, using the Kenya example.
2. Bond markets can lend cheaply for years and then turn suddenly, as in the UK in 2022. Can a government prepare for that, or only react?
3. Japan carries the rich world’s highest debt with the lowest borrowing costs. Is that a model other countries could follow, or a special case?
4. Compare fixed fiscal rules (the old euro-area 3%/60% limits) with country-specific spending paths. Which is more likely to work, and why?
5. You advise a government whose interest payments are rising toward a third of its revenue. Set out a three-step plan to bring the deficit down while limiting the political damage.

4.2.6. Source Notes

1. NIESR: “The Truss/Kwarteng Mini Budget: One Year On” <https://niesr.ac.uk/blog/truss-kwarteng-mini-budget-one-year>
2. Bank of England: “Bank of England announces gilt market operation” (28 September 2022) <https://www.bankofengland.co.uk/news/2022/september/bank-of-england-announces-gilt-market-operation>
3. MUFJ: “A Closer Look at Moody’s US AAA Downgrade” https://www.mufjamerica.com/sites/default/files/document/2025-06/Policy-Note_6_10_A-Closer-Look-at-Moodys-US-AAA-Downgrade.pdf
4. Peter G. Peterson Foundation: “With \$40 Trillion in Debt, Is the U.S. Headed for More Credit Downgrades?” <https://www.pgpf.org/article/with-40-trillion-in-debt-is-the-u-s-headed-for-more-credit-downgrades/>
5. European Commission: “Economic governance review” (reformed Stability and Growth Pact, 2024) https://economy-finance.ec.europa.eu/economic-and-fiscal-governance/economic-governance-review_en
6. France 24: “Kenya’s Ruto says finance bill to be withdrawn after anti-tax protest deaths” (26 June 2024) <https://www.france24.com/en/live-news/20240626-kenya-s-president-says-finance-bill-to-be-withdrawn-after-anti-tax-protest-deaths>

7. CNBC: UK PM Liz Truss fires Finance Minister Kwasi Kwarteng, 14 October 2022
<https://www.cnn.com/2022/10/14/uk-pm-liz-truss-fires-finance-minister-kwasi-kwarteng.html>
8. Federal Reserve Bank of St. Louis: Why Is Japan's Government Debt So High?, 2025
<https://www.stlouisfed.org/on-the-economy/2025/apr/what-is-behind-japan-high-government-debt>
9. Nippon.com: The Political History of Japan's Consumption Tax
<https://www.nippon.com/en/features/h00013/the-political-history-of-japan%E2%80%99s-consumption-tax.html>

4.3. The Business of Sport, Art and Culture

4.3.1. Goal

Contrast the parts of sport and culture that fund themselves commercially with the parts that depend on subsidy and philanthropy.

4.3.2. Reading

In a London auction room a single painting sells in under three minutes for more than a regional orchestra's entire annual budget. Both are "culture," but they belong to different economies. Sport, art and culture look like leisure; they are also industries, and they are funded in two very different ways.

At the commercial end, revenue comes from media rights, ticketing, sponsorship, player transfers and sales. In December 2023 the English Premier League sold its UK live-broadcast rights for a record GBP 6.7 billion (about US\$8.45 billion) over four years from 2025–26, the largest domestic sports-media deal in UK history^[1]. Media rights are top European football's single biggest revenue source, large enough that in 2023–24 Real Madrid became the first club ever to record more than EUR 1 billion of revenue in a year^[2].

That scale draws in outside capital with mixed motives. Saudi Arabia's Public Investment Fund, a sovereign wealth fund, bought Newcastle United in 2021 for about GBP 305 million, four Saudi domestic-league clubs in 2023, and reportedly around US\$5 billion into the breakaway LIV Golf circuit^[3]. It states three aims at once: a financial return, "soft power" (influence gained through visibility rather than force), and building a domestic entertainment industry.

The other model cannot cover its costs from tickets. US orchestras, on average, take more revenue from donations than ticket sales, and American concert tickets average close to US\$80, which limits audiences^[4]. Economist William Baumol

named the reason: a string quartet still needs four players for the same length of time as in 1800, so as wages rise economy-wide, the relative cost of live performance keeps climbing^[7]. This "cost disease" is why orchestras, opera and theatre need subsidy or philanthropy indefinitely. Countries differ sharply in how much public money they give — Germany funds a dense network of civic theatres and opera houses, the US very little, relying instead on tax-deductible donations.

The art market sits at the commercial end but is concentrated and, lately, shrinking. Global sales were an estimated US\$57.5 billion in 2024, down 12% for a second straight annual fall. The US took 43% of sales by value, the UK 18% and China 15%^[5].

Japan's cultural economy is lopsided in an instructive way. Its fine-art market is small, around 1% of world sales, and its orchestras and theatres rely on the same public-and-corporate mix as elsewhere^[8]. But one cultural industry is a genuine commercial powerhouse: animation. In 2023 the anime industry earned a record roughly JPY 3.3 trillion (about US\$21 billion), and for the first time in a full year, overseas revenue exceeded domestic^[6]. Anime works as a business the way a live orchestra cannot: it is reproduced at almost no extra cost, sold into every market at once, and earns from its characters for decades. The divide mirrors the topic's central split — culture that scales through reproduction can be a big business; culture performed live, each time, generally cannot pay for itself.

4.3.3. Vocabulary Focus

media / broadcast rights; sponsorship; sovereign wealth fund; “soft power”; philanthropy; “cost disease” (Baumol)

4.3.4. Reading Questions

1. What are the main revenue sources at the commercial end of sport and culture?
2. What three aims does Saudi Arabia’s Public Investment Fund state for its sports investments?
3. What is “cost disease,” and why does it affect live performance in particular?
4. How do Germany and the United States differ in how they fund the performing arts?
5. Why does the reading call anime a “commercial powerhouse” while Japan’s fine-art market is small?

4.3.5. Discussion Questions

1. Sovereign wealth funds now own major sports clubs and events. What are the benefits and risks of that kind of ownership for a league?
2. If “cost disease” means live performance will always need subsidy, is that a good reason to fund it publicly, or a reason to let ticket prices rise?
3. The reading says culture that “scales through reproduction” can be a big business. Does that mean live performance is destined to shrink? Give reasons.
4. Compare the funding of elite football and a public orchestra. Which model is more secure, and which produces better outcomes for audiences?
5. You advise a mid-sized city orchestra that covers only 40% of its costs from tickets. Propose a funding plan and explain the trade-offs in your choices.

4.3.6. Source Notes

1. SportsPro: “Premier League TV rights sold for record UK£6.7bn as Sky keeps lion’s share of domestic coverage” <https://www.sportspro.com/news/premier-league-tv-rights-deal-sky-tnt-amazon-2025-2029/>
2. Deloitte UK: “Real Madrid break billion Euro revenue record to top the Deloitte Football Money League” <https://www.deloitte.com/uk/en/about/press-room/deloitte-football-money-league.html>
3. Play the Game: “The Public Investment Fund’s growing sports portfolio” <https://www.playthegame.org/news/the-public-investment-fund-s-growing-sports-portfolio/>
4. Symphony.org (League of American Orchestras): “Radical Revenue” <https://symphony.org/features/radical-revenue/>
5. Art Basel: “Global art market sales reached an estimated \$57.5 billion in 2024” (Art Basel & UBS Global Art Market Report 2025) <https://www.artbasel.com/news/the-art-basel-and-ubs-global-art-market-report-2025>
6. Screen Daily: “Japanese anime industry generated record \$22bn in 2023 with overseas surpassing local takings” <https://www.screendaily.com/news/japanese-anime-industry-generated-record-22bn-in-2023-with-overseas-surpassing-local-takings/5198698.article>
7. Chicago Booth Review: Diagnosing William Baumol’s Cost Disease <https://www.chicagobooth.edu/review/diagnosing-william-baumols-cost-disease>

8. Artnet News: Japan Boasts the World's Third Largest Economy. So Why Does It Still Have a Disproportionately Small Share of the Global Art Market?, 2024 <https://news.artnet.com/market/japan-art-market-2205158>

4.4. Philanthropy, Foundations and Impact

4.4.1. Goal

Explain how large-scale institutional philanthropy is structured, why its accountability is questioned, and what the responses have been.

4.4.2. Reading

It is tempting to think of philanthropy as charity on a bigger scale — the same act, just more money. The structure is different, and the difference is what the debate is about. Charity is direct help given now. Modern institutional philanthropy is large sums held in an endowed foundation that gives away only a slice of its investment returns each year, backed by tax advantages, and increasingly judged on measurable “impact.”

That structure raises real questions. A private foundation can, in principle, exist forever: in the United States it must distribute only about 5% of its assets a year to keep its tax exemption. Donor-advised funds — accounts that give the donor an immediate tax deduction but let them recommend grants later — have no payout deadline at all^[1]. Critics, notably the political theorist Rob Reich, add that big philanthropy is subsidised by every taxpayer through deductions, is accountable to no electorate, and lets very wealthy individuals set public priorities in health, education and science^[7].

The responses try to answer those objections. One is the spend-down foundation, which commits to giving everything away and closing. In May 2025 the Gates Foundation said it would spend roughly US\$200 billion over the next 20 years, about double what it gave in its first 25, and shut permanently by the end of 2045 rather than last forever^[2].

A second response is unrestricted giving. Since 2019 MacKenzie Scott has given more than US\$19 billion to over 2,000 organisations, with no application process

and minimal reporting. A three-year study published in 2025 found the gifts were “transformational,” and that feared downsides — organisations overspending or losing other donors — had not materialised^[3].

A third response is the “impact” turn: formal metrics and impact investing, deploying capital for a measurable social return alongside a financial one, which had grown to an estimated US\$1.57 trillion in assets by 2024^[4]. A fourth is regulation: the United Kingdom requires charities to demonstrate “public benefit” and is home to one of the world’s largest foundations, the Wellcome Trust, whose roughly GBP 37.6 billion endowment funds health and science research^[5].

Japan’s philanthropic sector is small relative to its economy. Individual giving was around 0.23% of GDP in 2020, roughly a seventh of the US share, and large private foundations of the Gates or Wellcome type are rare^[6]. Three things explain this: tax incentives were weak until recent reforms; there is an expectation that the state and companies, not individuals, handle social provision; and until a 2008 reform, government ministries had wide discretion over which bodies served the public interest^[8]. Corporate giving is the stronger tradition, formalised in the Keidanren business federation’s “1% Club.”^[9] Recent large personal pledges by company founders suggest the picture may be shifting, but from a low base. The honest verdict on all these responses: partial answers, not a settled fix.

4.4.3. Vocabulary Focus

foundation; endowment; payout rule; donor-advised fund (DAF); spend-down (sunset)
foundation; impact investing

4.4.4. Reading Questions

1. What does the reading say separates institutional philanthropy from charity?
2. Why do critics argue that big philanthropy has an accountability problem?
3. What is a spend-down foundation, and what did the Gates Foundation announce in May 2025?
4. How does MacKenzie Scott's approach to giving differ from a traditional grant process?
5. Give two reasons the reading offers for why philanthropy is smaller in Japan than in the United States.

4.4.5. Discussion Questions

1. The reading calls the responses "partial answers, not a settled fix." Which response do you find most convincing, and why?
2. Should a foundation be allowed to exist forever while giving away only about 5% of its assets a year? Give reasons.
3. Unrestricted, no-strings giving was expected to cause problems that a 2025 study did not find. Why might large unrestricted gifts work better than targeted ones?
4. Compare tax-subsidised private giving with higher taxes and public spending as ways to fund health or education. What does each do better?
5. You advise a new foundation with US\$1 billion. Recommend a payout rate, whether to spend down or last indefinitely, and how you would measure success. Justify each choice.

4.4.6. Source Notes

1. The Regulatory Review (University of Pennsylvania): "Reforming Donor-Advised Funds" <https://www.theregreview.org/2024/04/20/reforming-donor-advised-funds/>
2. Gates Foundation: "Gates Foundation Will Double Spending Over Next 20 Years to Accelerate Progress on Saving and Improving Lives" (May 2025) <https://www.gatesfoundation.org/ideas/media-center/press-releases/2025/05/25th-anniversary-announcement>
3. Center for Effective Philanthropy: "Research Shows MacKenzie Scott's Large, Unrestricted Gifts Create Sustained Impact for Nonprofits and Communities" (2025) <https://cep.org/news/press-releases/research-shows-mackenzie-scotts-large-unrestricted-gifts-create-sustained-impact-for-nonprofits-and-communities/>
4. Global Impact Investing Network: "Sizing the Impact Investing Market 2024" <https://thegiin.org/publication/research/sizing-the-impact-investing-market-2024/>
5. Wellcome: "Wellcome annual report and financial statements 2024/25" (endowment ~GBP 37.6 billion at 30 September 2025) <https://wellcome.org/insights/articles/wellcome-annual-report-and-financial-statements-202425>
6. Alliance magazine: "Japanese philanthropy has fallen behind. Can it catch up?" <https://www.alliancemagazine.org/analysis/japanese-philanthropy-has-fallen-behind-can-it-catch-up/>

4.5. The Future of Money

4.5.1. Goal

Trace how money's form has changed over time and give a sober account of what is actually arriving next.

4.5.2. Reading

In Sweden, the share of people who used cash for their most recent purchase fell from about 40% in 2010 to about 8% in 2022^[1]. Germany moved the opposite way: cash was still used in 51% of everyday transactions in 2023, though only 26% of spending by value, since it dominates small purchases while cards take over larger ones^[2]. Same decade, same continent, opposite directions. “Money” has changed form many times before, and it is changing again.

Its form has moved in stages: commodity money, then minted coins, then paper notes — first backed by gold or silver, later “fiat” money that has value only because a government declares it legal tender. Then came book-entry bank deposits, payment cards, and mobile and QR-code payments. Each step kept the same unit of account, the krona or the yen, and changed only how value is stored and moved.

One point matters before looking ahead: most money is already digital and private. In modern economies the large majority of the money supply is commercial-bank deposits — money created when a bank lends, existing only as entries in a ledger — not central-bank cash^[3]. So “the future of money” is mostly a question about what backs and moves those deposits.

The newest forms sit on top of that. Privately issued tokens pegged to a national currency, over 90% to the US dollar, reached a combined value of about US\$250 billion by mid-2025^[4]. Reported transfer volume hit roughly US\$33 trillion

in 2025, though most is automated trading, not payments^[4]. Banks and central banks are separately piloting “tokenised deposits” on shared ledgers, so payment and settlement happen together, instantly^[5]. Central-bank digital currencies, a digital form of public central-bank money, are being tested in many countries, China and India furthest along, but most remain at the pilot stage^[8].

Sweden, having led the world toward cashlessness, is now legislating to protect cash. In November 2024 the authorities sent every household a brochure advising a week's worth of cash, around SEK 1,000 per adult, and backed a proposed legal duty for essential-goods shops to accept it^[6]. Cash is now treated as resilience infrastructure, a backup if the electronic system fails, rather than a relic.

Japan is the useful counter-example to the “cash is dying” story. QR-code wallets have spread quickly, yet Japan's demand for physical currency is around 19–20% of GDP, the highest the BIS tracks, and cash is still used in more than four in ten retail purchases^[7]. Several things hold it in place: near-zero rates removed any cost to holding cash over deposits; the population is old and used to tangible money; and crime is rare, so cash feels safe. Japan issued its first redesigned banknotes in twenty years in 2024, and only recently has cash in circulation begun to fall^[9]. Whether a society goes cashless depends on interest rates, demographics, trust and habit, not just available technology — why the same decade produced near-cashless Sweden and cash-heavy Japan.

4.5.3. Vocabulary Focus

fiat money; money supply; cashless society; stablecoin; tokenised deposit; central-bank digital currency (CBDC)

4.5.4. Reading Questions

1. How did cash use in Sweden change between 2010 and 2022?
2. What does the reading mean by saying “most money is already digital and private”?
3. Why does the reading say the reported US\$33 trillion stablecoin transfer figure overstates payment use?
4. What did Sweden do in November 2024, and how is cash now described there?
5. What four things does the reading say hold cash in place in Japan?

4.5.5. Discussion Questions

1. Sweden went furthest toward cashlessness and is now protecting cash. What does that reversal suggest about how fast countries should move?
2. If most money is already private bank deposits, how much would a central-bank digital currency actually change for an ordinary user?
3. Stablecoins are widely used for trading but less so for everyday payments. What would have to change for that to shift?
4. Compare Sweden and Japan. Which country's approach to the future of money looks wiser, and why?
5. You advise a central bank deciding whether to issue a digital currency. List the strongest argument for and against, and recommend a decision.

4.5.6. Source Notes

1. Sveriges Riksbank: “Payments Report 2024” <https://www.riksbank.se/en-gb/payments-cash/payments-in-sweden/payments-report-2024/>
2. Deutsche Bundesbank: “Payment behaviour in Germany in 2023” <https://www.bundesbank.de/en/press/press-releases/payment-behaviour-in-germany-in-2023-934894>
3. Bank of England: “Money creation in the modern economy” (Quarterly Bulletin, 2014 Q1) <https://www.bankofengland.co.uk/quarterly-bulletin/2014/q1/money-creation-in-the-modern-economy>
4. Bloomberg: “Stablecoin Transactions Rose to Record \$33 Trillion in 2025” <https://www.bloomberg.com/news/articles/2026-01-08/stablecoin-transactions-rose-to-record-33-trillion-led-by-usdc>
5. Bank for International Settlements: 2023 Annual Economic Report, Chapter III — “Blueprint for the future monetary system” (tokenisation and the unified ledger) <https://www.bis.org/publ/arpdf/ar2023e3.htm>
6. Cash Essentials: “Sweden Steps Up Efforts to Reverse the Decline in Cash Usage” <https://cashesentials.org/sweden-steps-up-efforts-to-reverse-the-decline-in-cash-usage/>
7. Bank of Japan: “Circulation Status of the New Series of Bank of Japan Notes” (July 2024 redesign; cash trends) https://www.boj.or.jp/en/research/wps_rev/rev_2025/data/rev25e09.pdf
8. Atlantic Council: “Central Bank Digital Currency Tracker” <https://www.atlanticcouncil.org/cbdctracker/>

9. The Japan Times: "Japan releases new banknotes for first time in 20 years", 3 July 2024
<https://www.japantimes.co.jp/news/2024/07/03/japan/society/new-banknotes-debut/>

Created September 17, 2026

Avanti Staff Corporation

